1. Write down the Difference Between [illegible]puter Architecture and Computer Organization.
2. Which type of instruction set architecture is used in modern processors such as Intel Core i7 — RISC. CISC, or a combination of both? Explain how it works.
3. Write one sentence Answer:

a)	Is decoding harder in RISC or in CISC?
b)	Is Intel processor fit or not into an AMD motherboard?
c)	What are the four basic steps of CPU operation?
d)	Match the processors in Set A = {Core i3, i5, i7, i9} with their correct performance categories in Set B = {Entry-level, Gaming & creative performance. Enthusiast & high-end performance, Balanced performance}.

1. Let's say you're uploading a 2 GB video file from your laptop to cloud storage (like Google Drive or AWS S3) with Transfer rate (R) = 50 Mbps. Average access latency (TA) = 120 ms for Time for handshake, connection setup, DNS lookup, and request acknowledgment, then calculate average time to read and write. 3
2. Show the memory hierarchy and Draw the single and three-level cache organization. 6
3. Derive the equation of two-level cache memory by using probability flowchart. 4
4. If the main memory access time is 100 ns, cache access time is 50 ns, cache hit rate is 90%, then what is AMAT to read from memory? 4
5. Briefly explain the memory access methods in computer systems with suitable examples. 3